Project Report on

ENTERPRISE NETWORK DESIGN FOR A TRADING FOR SUPPORT CENTRE

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BONAFIDE CERTIFICATE

Certified that this project report **Enterprise Network Design for a Trading Floor Support Centre** is the bonafide work of **JYOTI RAJPUT** who carried out the project work under my/our supervision

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TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION ……6

1.1 Identification of Client / Need / Relevant Contemporary Issue

1.2 Identification of Problem

1.3 Identification of Tasks

1.4 Timeline

1.5 Organization of the Report

CHAPTER 2: LITERATURE REVIEW / BACKGROUND STUDY ……7

2.1 Timeline of the Reported Problem

2.2 Existing Solutions

2.3 Bibliometric Analysis

2.4 Review Summary

2.5 Problem Definition

2.6 Goals / Objectives

CHAPTER 3: DESIGN FLOW / PROCESS ……8-9

3.1 Evaluation & Selection of Specifications / Features

3.2 Design Constraints

3.3 Analysis of Features and Finalization Subject to Constraints

3.4 Design Flow

3.5 Design Selection

3.6 Implementation Plan / Methodology

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION……10-12

4.1 Implementation of Solution

CHAPTER 5: CONCLUSION AND FUTURE WORK ……..……..13

5.1 Conclusion

5.2 Future Work

REFERENCES ……..…….13

APPENDIX ……14-16

1 Plagiarism Report

2 Design Checklist

TITLE OF PROJECT REPORT

Enterprise Network Design for a Trading Floor Support Centre Using VLANs, OSPF, and Dual ISP Redundancy

ABSTRACT

The rapid growth of organizations and increasing dependence on digital communication systems have made efficient network design a critical requirement. This project focuses on designing and implementing a robust, scalable, and secure network infrastructure for a Trading Floor Support Centre consisting of 600 employees relocating to a new building. The proposed solution is developed using Cisco Packet Tracer and follows a hierarchical network model to ensure efficient management, performance, and scalability.

The network is logically segmented using Virtual Local Area Networks (VLANs), where each department is assigned a separate VLAN and subnet to enhance security and reduce broadcast traffic. Inter-VLAN communication is achieved using multilayer switches, while Open Shortest Path First (OSPF) routing protocol is implemented to ensure dynamic and efficient route exchange. Redundancy is achieved by connecting the network to multiple Internet Service Providers (ISPs) through dual routers, ensuring high availability and minimizing downtime.

Dynamic Host Configuration Protocol (DHCP) is used for automatic IP address allocation, while static IP addresses are assigned to critical devices in the server room. Network Address Translation (NAT) is configured to enable internet access using public IP addresses. Security features such as Secure Shell (SSH), Access Control Lists (ACLs), and port security are implemented to protect the network from unauthorized access.

The implementation and testing results demonstrate successful communication within VLANs, efficient routing, secure access, and reliable redundancy. The designed network meets current organizational requirements and provides a strong foundation for future expansion and advanced networking solutions.

CHAPTER 1: INTRODUCTION

1.1 Identification of Client / Need / Relevant Contemporary Issue

The client for this project is a **Trading Floor Support Centre** with approximately 600 employees that is relocating to a new building without any existing network infrastructure. In today's digital era, organizations rely heavily on efficient and secure networking systems to support communication, data sharing, and business operations. Therefore, there is a critical need to design a reliable and scalable network that meets both current operational requirements and future expansion.

A major contemporary issue addressed in this project is the demand for **high availability and redundancy**. In business environments such as trading floors, network downtime can lead to significant financial and operational losses. Hence, the network must ensure uninterrupted connectivity through redundant paths and multiple Internet Service Providers (ISPs). Additionally, the increasing need for **wireless connectivity** requires each department to have dedicated wireless networks for seamless mobility.

1.2 Identification of Problem

The primary problem is the absence of a structured network in the new building. Without proper network design, communication between departments would be inefficient, insecure, and prone to congestion. The challenge lies in designing a network that supports multiple departments, ensures data security, provides internet access, and maintains high performance for all users.

1.3 Identification of Tasks

To address the problem, several tasks must be performed, including designing the network topology, configuring routers and switches, implementing VLANs for segmentation, setting up routing protocols like OSPF, enabling security features, and testing connectivity and redundancy.

1.4 Timeline

The project is planned over multiple phases including design, implementation, testing, and documentation, ensuring systematic completion within the given timeframe.

1.5 Organization of the Report

This report is structured into five chapters covering introduction, literature review, design methodology, implementation, and conclusion with future scope.

CHAPTER 2: LITERATURE REVIEW / BACKGROUND STUDY

2.1 Timeline of the Reported Problem

In early networking environments, organizations used flat network structures where all devices were connected within a single broadcast domain. As the number of users increased, issues such as network congestion, broadcast storms, and security vulnerabilities became more prominent. To address these challenges, technologies like VLAN (Virtual Local Area Network) were introduced, enabling logical segmentation of networks. Over time, routing protocols such as OSPF (Open Shortest Path First) and techniques like NAT (Network Address Translation) further improved network scalability and efficiency.

2.2 Existing Solutions

Several technologies have been developed to overcome networking challenges. VLANs are used to segment networks into smaller, manageable units, improving performance and security. Subnetting helps in efficient IP address allocation. Routing protocols like OSPF ensure dynamic and efficient routing between networks. NAT allows multiple devices to share a single public IP address for internet access. Additionally, security mechanisms such as Access Control Lists (ACLs) and port security are implemented to restrict unauthorized access.

2.3 Bibliometric Analysis

Different technologies offer distinct advantages. VLANs reduce broadcast traffic but require routing for inter-VLAN communication. OSPF provides fast convergence and scalability but involves complex configuration. NAT conserves IP addresses but introduces processing overhead. Security techniques like ACLs and port security enhance protection but require careful planning and management.

2.4 Review Summary

The literature suggests that no single technology can solve all networking issues. A combination of VLANs, routing protocols, NAT, and security measures is essential for building an efficient and secure network infrastructure.

2.5 Problem Definition

The problem is to design a network that supports multiple departments, ensures secure communication, provides redundancy, and efficiently manages resources.

2.6 Goals / Objectives

The main objectives include achieving network scalability, ensuring security, providing reliable connectivity, and enabling efficient communication across departments.

CHAPTER 3: DESIGN FLOW / PROCESS

3.1 Evaluation & Selection of Specifications / Features

The network design for the Trading Floor Support Centre requires careful selection of features to ensure efficiency, scalability, and security. Key features selected include VLAN segmentation for separating departmental traffic, OSPF as the routing protocol for dynamic route exchange, dual ISP connectivity for redundancy, and wireless access points for mobility. Additionally, DHCP is used for automatic IP address allocation, while static IP addressing is reserved for critical devices such as servers.

3.2 Design Constraints

Several constraints influence the network design. These include hardware limitations such as the number of available switch ports and router interfaces, budget considerations, and the need to ensure high security standards. The design must also comply with networking standards and support future scalability without major restructuring.

3.3 Analysis of Features and Finalization Subject to Constraints

After evaluating the features and constraints, the final design incorporates VLAN-based segmentation to reduce broadcast domains and improve performance. OSPF is selected due to its scalability and fast convergence. Redundancy is achieved through dual routers and multiple ISP connections. Security measures such as port security and ACLs are included to protect sensitive departments like Finance.

3.4 Design Flow

The design process follows a structured approach:

1. Develop network topology
2. Assign VLANs and subnets
3. Configure routing using OSPF
4. Implement DHCP and IP addressing
5. Configure security mechanisms
6. Test and validate network performance

3.5 Design Selection

A hierarchical network model is chosen, consisting of core, distribution, and access layers. This design simplifies management, improves performance, and supports scalability.

3.6 Implementation Plan / Methodology

The implementation involves configuring routers, multilayer switches, and access switches in Cisco Packet Tracer. VLANs are created for each department, routing is enabled using

multilayer switches, and redundancy is ensured through dual ISP connections. Security configurations and testing are performed to validate the network.

CHAPTER 4: RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

The designed network was successfully implemented using Cisco Packet Tracer by configuring routers, multilayer switches, and access switches according to the requirements. The implementation included VLAN creation, inter-VLAN routing, DHCP configuration, OSPF routing, NAT, and security mechanisms.

Basic Device Configuration

All devices were configured with hostnames, passwords, and security settings:

```
enable
configure terminal
hostname R1

enable secret admin123
line console 0
password admin123
login
logging synchronous
exec-timeout 2 0

line vty 0 4
password admin123
login
transport input ssh

service password-encryption
no ip domain-lookup
banner motd #UNAUTHORIZED ACCESS PROHIBITED#
```

VLAN Configuration (Multilayer Switch)

```
vlan 10
name SALES
vlan 20
name HR
vlan 30
name FINANCE
vlan 40
name ADMIN
vlan 50
name ICT
```

Assigning VLANs to ports:

```
interface range fa0/1-10
switchport mode access
switchport access vlan 10
```

Inter-VLAN Routing

```
interface vlan 10
ip address 172.16.10.1 255.255.255.0
no shutdown

ip routing
```

DHCP Configuration

```
ip dhcp pool SALES
network 172.16.10.0 255.255.255.0
default-router 172.16.10.1
```

OSPF Routing

```
router ospf 1
network 172.16.0.0 0.0.255.255 area 0
```

NAT Configuration (Router)

```
access-list 1 permit 172.16.0.0 0.0.255.255

interface g0/0
ip nat inside

interface g0/1
ip nat outside

ip nat inside source list 1 interface g0/1 overload
```

Port Security (Finance Department)

```
interface fa0/5
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation shutdown
```

Testing and Validation

- Devices within same VLAN communicated successfully

- Inter-VLAN routing worked correctly
- Internet access via NAT was successful
- OSPF routes were properly advertised
- Port security blocked unauthorized devices

Verification Commands

```
show ip route
show vlan brief
show ip dhcp binding
show port-security
ping 8.8.8.8
```

Result Summary

Test Case	Result
VLAN Communication	Successful
Inter-VLAN Routing	Successful
Internet Access	Successful
Security Enforcement	Successful

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

The project successfully achieved its objective of designing and implementing a robust, scalable, and secure network infrastructure for the Trading Floor Support Centre. By adopting a hierarchical network model, the design ensures efficient traffic management, simplified troubleshooting, and future scalability. The use of VLANs effectively segmented the network into different departments, reducing broadcast traffic and enhancing security.

The implementation of inter-VLAN routing using multilayer switches enabled seamless communication between departments where required. The use of OSPF as a dynamic routing protocol ensured efficient route exchange and quick convergence in case of network changes. Redundancy was achieved by connecting the network to multiple ISPs through dual routers, ensuring high availability and minimizing downtime.

Security was a key focus of the design. Features such as SSH for secure remote access, port security in sensitive departments, and ACLs for traffic control significantly improved the overall network protection. Additionally, NAT configuration enabled efficient internet access while conserving public IP addresses.

The testing and validation phase confirmed that all components functioned as expected. Devices within the same VLAN communicated successfully, inter-VLAN routing was operational, and redundancy mechanisms worked effectively. Overall, the network design meets current organizational requirements while providing a strong foundation for future expansion.

5.2 Future Work

Although the current network design meets all specified requirements, several enhancements can be implemented in the future to further improve performance and security. Advanced security features such as firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS) can be integrated to protect against sophisticated cyber threats.

The network can also be upgraded to support cloud-based services and virtualization technologies, enabling better resource utilization and scalability. Implementation of network monitoring tools and automation using software-defined networking (SDN) can improve management efficiency and reduce manual intervention.

Additionally, IPv6 deployment can be considered to address future IP address requirements. Wireless networks can be enhanced using advanced authentication mechanisms such as WPA3 and centralized wireless controllers.

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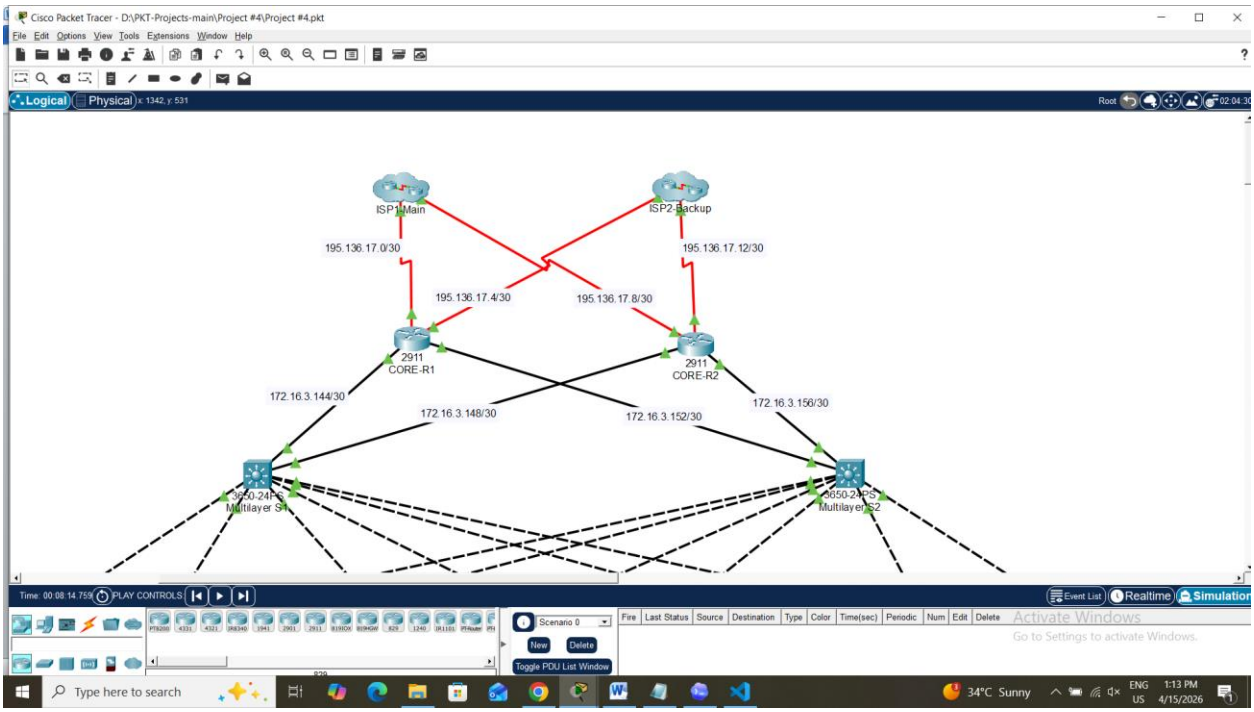
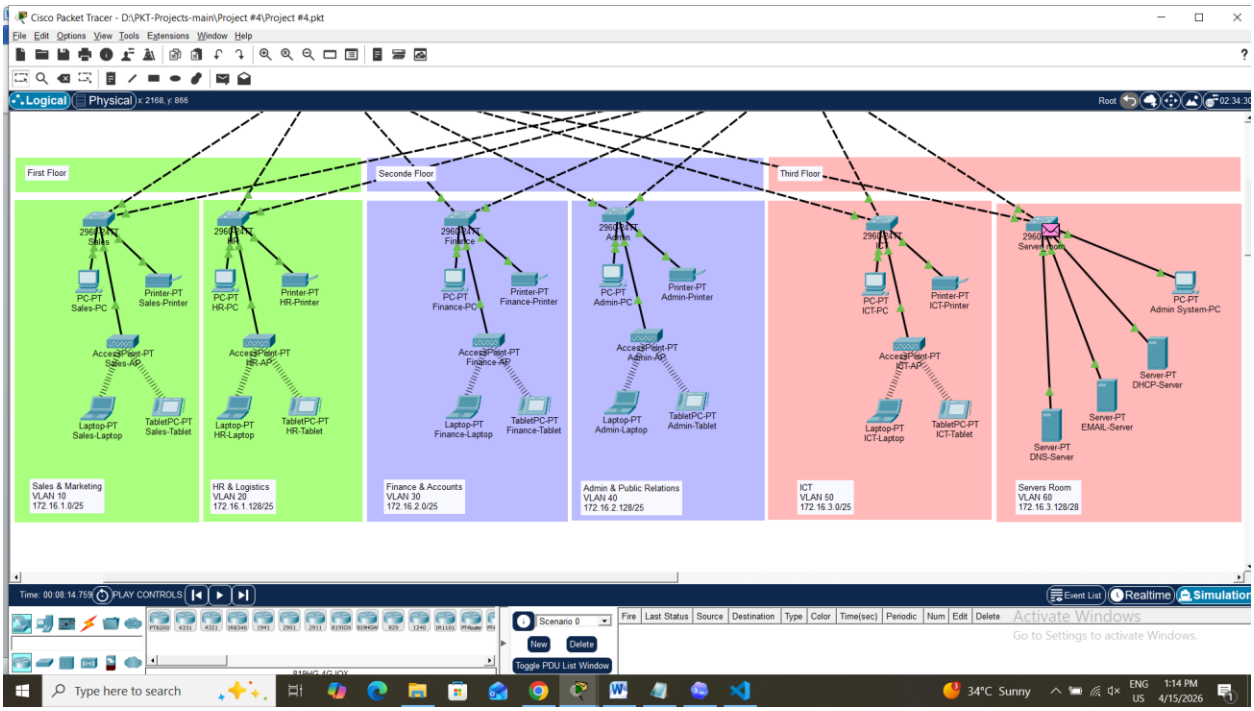
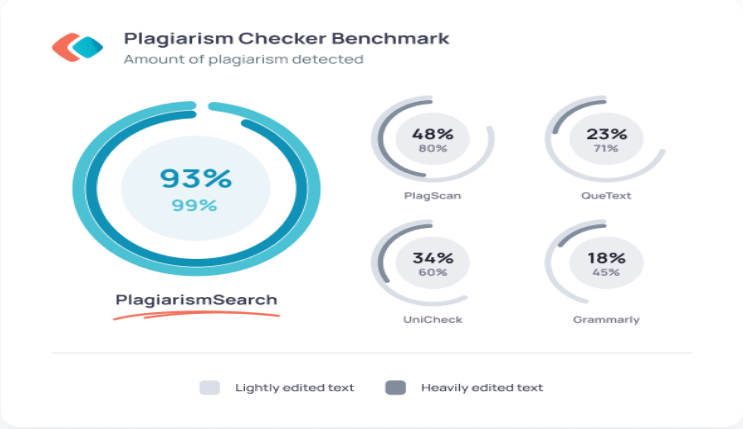
APPENDIX

Screenshots :





A.1 Plagarism Report



A.2 Design Checklist

Network topology designed using hierarchical model (Core, Distribution, Access)

✔ All devices (routers, switches, APs) correctly placed and connected

✔ Redundant links and dual ISP connectivity configured

✔ VLANs created for each department

✔ Proper subnetting applied (172.16.1.0 network)

✔ Inter-VLAN routing configured on multilayer switches
✔ OSPF routing protocol implemented and verified
✔ DHCP configured for dynamic IP allocation
✔ Static IP addresses assigned to server room devices
✔ NAT (PAT) configured for internet access
✔ Wireless networks configured for each department
✔ SSH enabled for secure remote access
✔ Port security configured (Finance department)
✔ ACLs implemented for traffic control
✔ Network tested for connectivity, redundancy, and security